LOTUS 260413: organism order — taxonomy vs NPClassifier class profiles
(taxa with ≥ 100 class-labelled molecules, at most the 60 richest)
60 taxa · Mantel r (Spearman) = 0.75, p = 0.001 (999 permutations) · cophenetic correlation between trees = 0.83

Open Tree of Life taxonomy

chemical-profile clustering

Asterales
Lamiales
Fabales
Sapindales
Malpighiales
Gentianales
Rosales
Asparagales
Apiales
Ranunculales
Solanales
Caryophyllales
Ericales
Magnoliales
Myrtales
Malvales
Poales
Brassicales
Laurales
Celastrales
Zingiberales
Saxifragales
Piperales
Fagales
Dipsacales
Cucurbitales
Austrobaileyales
Liliales
Santalales
Boraginales
Alismatales
Vitales
Cupressales
Pinales
Araucariales
Polypodiales
Jungermanniales
Florideophycidae [no order]
Eurotiales
Hypocreales
Xylariales
Sordariales
Diaporthales
Pleosporales
Lecanorales
Polyporales
Agaricales
Russulales
Dictyoceratida
Haplosclerida
Axinellida
Tetractinellida
Alcyonacea
Primates
Enterogona
Rhabditida
Streptomycetales
Cyanobacteria [no order]
Enterobacteriales
Bacillales

Gentianales
Asparagales
Liliales
Ranunculales
Magnoliales
Laurales
Piperales
Sapindales
Apiales
Asterales
Zingiberales
Lamiales
Dipsacales
Solanales
Poales
Brassicales
Alismatales
Fabales
Rosales
Malvales
Caryophyllales
Ericales
Myrtales
Fagales
Saxifragales
Vitales
Boraginales
Santalales
Malpighiales
Polypodiales
Cucurbitales
Cupressales
Pinales
Araucariales
Jungermanniales
Celastrales
Austrobaileyales
Eurotiales
Hypocreales
Xylariales
Pleosporales
Diaporthales
Sordariales
Polyporales
Agaricales
Russulales
Lecanorales
Florideophycidae [no order]
Dictyoceratida
Haplosclerida
Axinellida
Enterogona
Primates
Rhabditida
Enterobacteriales
Tetractinellida
Cyanobacteria [no order]
Bacillales
Streptomycetales
Alcyonacea

5 4 3 2 1 0
taxonomic distance (ranks)

0.0 0.2 0.4 0.6 0.8
Bray–Curtis distance of class profiles (average linkage)

Archaeplastida
Fungi
Metazoa
Bacteria